PlungerSet-80 Automated Stoppering Station

PlungerSet Series · Elara Automation GmbH

Lifecycle Information Document

The PlungerSet-80 is a benchtop automated stoppering station that inserts rubber plungers into pre-filled syringes in laboratory and pilot pharmaceutical filling lines. This document covers the full operational lifecycle of the product, from pre-installation planning through to end-of-life disposal.

It should be read alongside the OPC UA NodeSet file (EA-NS-PS080.NodeSet2.xml), which provides the machine-readable interface definition for SCADA and MES integration, and the line description document (EA-LN-SPL01-EN) for multi-station context.

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About This Document

This document has been prepared by Elara Automation GmbH and is intended for qualified technical personnel responsible for the installation, commissioning, operation, and maintenance of the equipment described herein. Readers are expected to be familiar with general principles of industrial automation, electrical safety, and network-connected laboratory equipment.

The document is structured as a reference manual. Sections are broadly ordered to follow the typical lifecycle of the equipment — from first delivery through to end-of-life disposal — but individual sections may be read in isolation. Cross-references are provided where information in one section depends on material covered elsewhere.

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Conventions Used in This Document

The following conventions are used throughout:

Notation	Meaning
{serialNumber}	The unit's serial number as printed on the rear label (e.g. EA-PS-2024-00391)
{base}	The site-specific topic or namespace prefix, configured during commissioning
NOTE	Supplementary information that aids understanding but is not safety-critical
CAUTION	Risk of equipment damage or process disruption if instruction is not followed
WARNING	Risk of personal injury if instruction is not followed

Related Documents

Document No.	Title
EA-NS-PS080.NodeSet2.xml	PlungerSet-80 OPC UA NodeSet (address space definition)
EA-LN-SPL01-EN	SPL-01 Line Description & Bill of Materials
EA-LC-LF120-EN	LinFill-120 Lifecycle Information Document (shared platform)

1 Product Overview

The PlungerSet-80 combines three mechanisms in a single compact unit. A vertical lift — mechanically identical to that of the LinFill-120 filling station — positions the stoppering head at the correct height above the syringe. A triangular-profile magazine channel holds up to four rubber plungers in vertical orientation; the triangular cross-section was selected after design testing of circular and semicircular profiles as it minimises inter-plunger friction while maintaining alignment. An SG90 servo motor drives a custom screw mechanism to advance one plunger at a time into the insertion chamber, from which a 12 V linear actuator executes the insertion stroke.

The station communicates via OPC UA (Unified Architecture), a platform-independent, service-oriented communication protocol widely adopted in pharmaceutical and process automation. Unlike the MQTT interface used on the LinFill-120, the OPC UA server embedded in the PlungerSet-80 exposes a structured address space with typed variables, callable methods, and standardised companion-specification objects that can be browsed and consumed directly by any OPC UA client without a separate interface document.

The PlungerSet-80 shares its base electronics platform (ESP32, L298N, firmware baseline 1.4.x) and lead-screw assembly with the LinFill-120. Spare parts for the lift assembly are interchangeable between the two product lines. The stoppering-specific sub-assemblies — magazine, servo, gear drive, and linear actuator — are unique to the PlungerSet-80.

Compatible Plungers

The station is tested and validated with 1 mL ISO standard rubber plungers. Validated types include West Pharmaceutical 4432/50 Gray and Datwyler FM27 series. Other plunger types may be compatible but should be evaluated by the equipment owner before use. Plunger consumables are not supplied by Elara Automation and must be sourced separately.

2 Delivery and Pre-Installation

Inspect the shipping carton for transit damage before opening. The package contains the PlungerSet-80 station (1×), the AC/DC power adapter for the ordered region, M6 mounting hardware (1 set), a printed quick-start card, and a pre-loaded test magazine with four dummy plungers for initial commissioning. The production plunger supply is not included.

The serial number is printed on a label affixed to the rear panel and on the packing slip. Record it in the site asset register immediately. The serial number is in the format EA-PS-YYYY-NNNNN (e.g. EA-PS-2024-00391) and is used as part of the OPC UA server endpoint identifier.

For short-term storage before installation, maintain the unit in its original packaging at temperatures between –20 °C and +60 °C in a dry environment. The PLA magazine channel is dimensionally stable up to 55 °C; do not store the unit near heat sources. For long-term storage exceeding three months, remove the plunger magazine and store it separately in a sealed bag.

3 Installation

Installation requires a qualified automation engineer familiar with OPC UA certificate management and network configuration. The steps below must be completed in the order given; attempting OPC UA connection before mechanical installation and power-on is complete may produce misleading diagnostic information.

3.1 Mechanical Mounting

Mount the station on an ACOPOS-compatible table or equivalent using four M6 bolts on the 150 × 120 mm bolt-circle pattern. Tighten to 8 N·m on a clean, flat surface. The table must be level to ±0.5 mm across the footprint. Maintain a minimum clearance of 120 mm above the stoppering head in the fully extended position. Overall dimensions are 240 mm (W) × 200 mm (D) × 480 mm (H) with the lift extended, 330 mm (H) retracted. Unit mass is 3.1 kg with the standard four-plunger magazine.

WARNING: The linear actuator extension force is rated at 50 N, which is sufficient to cause injury. Ensure the stoppering chamber is clear before issuing a Start method call. Never reach into the magazine slot while the station is powered.

3.2 Electrical Connection

Connect the included power adapter to the 5.5/2.1 mm barrel jack (centre positive, 12 V DC, 3 A minimum). The standard-magazine configuration draws up to 22 W during simultaneous actuation of all mechanisms. If the extended magazine kit (EA-PS080-MAG-EXT) is fitted, peak draw increases to approximately 28 W; verify the power supply rating before installing optional kits.

The SG90 servo is powered from the L298N onboard 5 V regulator; the lift motor and linear actuator are driven from the 12 V rail. A 3.15 A slow-blow fuse (rear panel) protects the 12 V supply. Replace only with an identical rating.

3.3 Network and OPC UA Configuration

The embedded OPC UA server listens on TCP port 4840 via the ESP32 WiFi interface. Connect the device to the facility 2.4 GHz WiFi network using the provisioning web interface (open the "PlungerSet-Setup" access point on first power-on, navigate to 192.168.4.1, enter the SSID and passphrase). Assign a static IP or DHCP reservation so that OPC UA clients can reach the device consistently.

After WiFi commissioning, connect an OPC UA client and browse to the server endpoint **opc.tcp://plungerset80-{serialNumber}.local:4840**. The server supports None, Sign, and SignAndEncrypt security modes. For production use, configure SignAndEncrypt with mutual X.509 certificate authentication. Exchange the server certificate (exportable from the provisioning UI) and the client certificate during commissioning.

The OPC UA address space is fully described in the NodeSet file EA-NS-PS080.NodeSet2.xml, which can be imported into SCADA, MES, or DCS tools to automatically populate the tag database. The namespace URI is http://elara-automation.de/UA/PlungerSet80/ at namespace index 1.

4 Commissioning and Qualification

Commissioning verifies that the station is correctly installed and communicating before the first production use. The steps below represent the minimum recommended by Elara Automation; additional OQ steps may be required by site quality procedures.

Step	Action	Expected Result
C-01	Power on; LED illuminates	Power LED steady green within 5 s
C-02	Connect OPC UA client; browse address space	PlungerSet80 root object visible; State node reads "ERROR" (pre-home)
C-03	Read FirmwareVersion node (ns=1;s=PlungerSet80.FirmwareVersion)	Returns "1.4.2" or later
C-04	Call Home method	State transitions ERROR/IDLE → HOMING → IDLE; no fault
C-05	Load test magazine; call Start	State IDLE → RUNNING → IDLE; CycleTime node updated; MagazineCount decremented

Step	Action	Expected Result
C-06	Call Start; immediately call Stop	State returns to IDLE; CycleTime not updated; MagazineCount not decremented
C-07	Verify ErrorCode node reads 0x0000 after successful Home	ErrorCode == 0
C-08	Induce magazine-empty fault (MagazineCount = 0); call Reset	State ERROR → IDLE; ErrorCode clears to 0x0000

4.1 OPC UA Session Watchdog

The firmware implements a configurable session watchdog: if the OPC UA session is inactive for longer than the watchdog timeout (default 120 seconds), the station transitions to ERROR with ErrorCode 0x0010 (COMMS_WATCHDOG). This behaviour prevents the station from remaining in an unknown state if the supervisory client disconnects unexpectedly. Integrators should configure the OPC UA client session keep-alive interval to less than 60 seconds to prevent spurious watchdog faults during normal operation.

5 Normal Operation

During production the PlungerSet-80 is controlled by calling OPC UA methods on the PlungerSet80 object in namespace index 1. The four methods — Start, Stop, Home, and Reset — correspond directly to the four command payloads of the LinFill-120 MQTT interface, enabling a supervisory system to use a common operational pattern across both station types.

The State variable (ns=1;s=PlungerSet80.State) is the primary operational indicator and should be monitored via an OPC UA subscription. The recommended subscription publishing interval is 500 ms or less. The MagazineCount variable (ns=1;s=PlungerSet80.MagazineCount) is decremented after each successful insertion and should be monitored to prompt the operator to refill the magazine. When MagazineCount reaches zero, a further Start call will result in an ERROR state with ErrorCode 0x0008 (MAGAZINE_EMPTY).

NOTE: *The Home method must be called at least once after each power-on before issuing a Start command. The station powers on in ERROR state (ErrorCode 0x0001, HOMING_TIMEOUT) until the first successful homing cycle.*

5.1 Error Codes

The ErrorCode variable carries a bitmask of active faults. Multiple faults may be present simultaneously; the value is the bitwise OR of all active codes.

Code (hex)	Name	Meaning	Recommended Action
0x0001	HOMING_TIMEOUT	Lift did not reach limit switch within 15 s	Check for mechanical obstruction; verify limit switch continuity
0x0002	INSERTION_FAULT	Linear actuator did not complete stroke within 5 s	Check for plunger jam; inspect actuator end-stop; issue Home after Reset
0x0004	SERVO_STALL	SG90 servo overcurrent detected	Remove obstruction from magazine feed path; inspect gear mesh
0x0008	MAGAZINE_EMPTY	MagazineCount == 0	Reload magazine; write correct count to MagazineCount node; issue Reset
0x0010	COMMS_WATCHDOG	OPC UA session inactive beyond watchdog timeout (default 120 s)	Restore OPC UA client connection; reduce client keep-alive interval
0x0020	ACTUATOR_RETRACT	Linear actuator failed to retract within 4 s	Check cable routing; inspect actuator for mechanical damage; issue Home after Reset

6 Preventive Maintenance

Perform the following tasks every 500 operating hours or at least once per year, whichever is sooner. Isolate power before accessing any moving components.

WARNING: Always isolate power before accessing the lead screw, linear actuator, servo gear drive, or magazine assembly. The 12 V supply is present at the L298N terminals even when the OPC UA server is not commanding motion.

Task	Procedure	Acceptance Criterion
Lead-screw lubrication	Apply NLGI #2 grease along full travel; manually traverse carriage; wipe excess.	Carriage moves without binding; no metal-on-metal contact sound
Magazine channel inspection	Remove magazine; inspect PLA channel for cracks, deformation, or contamination. Clean with dry wipe. Replace if any crack is found (EA-PS080-SPARE-MAG).	No cracks; plungers slide freely without snagging
Servo gear mesh check	Manually rotate servo output and assess gear mesh. Replace gear set if backlash at the screw exceeds 1 mm (EA-PS080-SPARE-SRV).	Backlash at screw tip < 1 mm; no audible gear wear
Linear actuator inspection	Verify 20 mm stroke by calling Start with empty magazine (manually block with plunger removed). Listen for end-stop contact. Inspect cable routing.	Full stroke reached; no unusual noise; cable not kinked
Fastener check	Check all M3 screws on motor mount, switch mount, gear housing, and frame brackets for tightness at 0.5 N·m.	No loose fasteners

7 Firmware Updates

Firmware updates are shared with the LinFill-120 platform but must only be applied using the PlungerSet-80 specific .bin file. Applying a LinFill-120 firmware image to this unit will result in incorrect servo and actuator behaviour.

- Connect host computer to the same WiFi network as the station.
- Navigate to **http://plungerset80-{serialNumber}.local** and log in.
- Select **System > Firmware Update**; upload the .bin file.
- Do not power off during the update (~45 s).
- After reboot, confirm FirmwareVersion node value in an OPC UA client.
- Call Home and verify the station returns to IDLE.
- Record firmware version and date in the maintenance log.

8 Spare Parts and Accessories

Use only genuine Elara Automation replacement parts. The magazine channel (EA-PS080-SPARE-MAG) requires firmware ≥ 1.3; the SG90 servo replacement (EA-PS080-SPARE-SRV) is pre-assembled with the gear set and requires firmware ≥ 1.4.2 to detect the replacement via the servo stall monitor.

Order Code	Description	Replaceability
EA-PS080-SPARE-ACT	12 V linear insert actuator assembly, 20 mm stroke	Authorised service personnel; requalification required

Order Code	Description	Replaceability
EA-PS080-SPARE-SRV	SG90 servo with pre-mounted gear assembly	User-replaceable; requires firmware ≥ 1.4.2
EA-PS080-SPARE-MAG	Triangular magazine channel, 4-plunger capacity, PLA	User-replaceable; requires firmware ≥ 1.3
EA-PS080-MAG-EXT	Extended magazine kit, 8-plunger capacity	User-installable; verify power supply ≥ 3 A at 28 W peak
EA-PS080-SVC-1Y	1-year extended service contract, 48 h on-site response	Service contract

9 Decommissioning and Disposal

When the PlungerSet-80 is taken permanently out of service, complete the following steps before disposal. Issue a decommissioning record quoting the serial number, date, and disposal route.

- Call the Home method; confirm State reads IDLE. Power off and remove the barrel-jack connector.
- Remove the plunger magazine and dispose of any remaining rubber plungers per site SOPs.
- Revoke the OPC UA server certificate in the provisioning UI to prevent unintended network connections from the device in the future.
- Perform a factory reset (hold rear-panel reset button 10 s until LED flashes amber) to erase WiFi credentials and network configuration.
- Disassemble for recycling: powder-coated steel base as ferrous metal; PLA magazine as industrial compostable waste; PCBs (ESP32, L298N) as WEEE per Directive 2012/19/EU.

10 Certifications and Regulatory Compliance

- CE marking: Low Voltage Directive (LVD) 2014/35/EU
- CE marking: EMC Directive 2014/30/EU
- CE marking: RoHS Directive 2011/65/EU
- WiFi radio — ESP32-WROOM-32 module FCC ID: 2AC7Z-ESPWROOM32
- IP Rating: IP20 · Pollution Degree 2 · Indoor use only

11 Warranty

24 months from date of shipment for manufacturing defects. Exclusions: magazine channel wear, SG90 servo damage resulting from overload or magazine obstruction, 3D-printed structural parts, plunger consumables, and damage caused by operation outside the electrical or environmental specifications stated in this document. Unauthorised disassembly voids the warranty.

Manufacturer & Technical Support	Sales & Ordering
Elara Automation GmbH Brückenstraße 14 70173 Stuttgart Germany Tel: +49 711 4920 0 E-mail: support@elara-automation.de Web: www.elara-automation.de	For orders and pricing, quote the order code printed on the unit label. When raising a support request, please provide: · Serial number (format EA-PS-YYYY-NNNNN, rear label) · Firmware version (FirmwareVersion OPC UA node) · OPC UA client software name and version · Description of observed behaviour and ErrorCode value E-mail: sales@elara-automation.de